INCH-POUND
MIL-C-24770(SH)
20 March 1992

MILITARY SPECIFICATION

CARTRIDGES, BROMINE, FOR POTABLE WATER DISINFECTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two sizes of flow through, water treatment cartridges which contain ion exchange resin impregnated with elemental bromine. The cartridges and associated equipment are used to disinfect potable water.

1.2 Classification. Bromine cartridges are of the following types as specified (see 6.2):

Type	Description
I	Small diameter (3-7/8 inch) cartridge.
II	Large diameter (7-13/16 inch) cartridge

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4610
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SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-T-60 - Tape: Packaging, Waterproof.

MILITARY

MIL-P-116 - Presentation, Methods of.

STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and Publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Code of Federal Regulations, Title 40, Part 22 - Federal Insecticide, Fungicide, and Rodenticide Act.

Code of Federal Regulations, Title 40, Part 141 - National Primary Drinking Water Regulations (NOTAU).

Code of Federal Regulations, Title 40, Part 143 - National Secondary Drinking Water Regulations (NOTAU).

(The Code of Federal Regulations is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. When indicated, reprints of certain regulations may be obtained from the Federal Agency responsible for issuance of the specific document.)

2.2 Non-Government Publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN PUBLIC HEALTH ASSOCIATION (APHA)
Standard Methods for the Examination of Water and Wastewater.

(Application for copies should be addressed to the American Public Health Association, 1015 Fifteenth St., NW, Washington, DC 20005.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 1193 - Standard Specification for Reagent Water. (DoD adopted)
D 1248 - Standard Specification for Polyethylene Plastics Molding and Extrusion Materials. ‘(’DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL SANITATION FOUNDATION (NSF)
60 - Drinking Water Treatment Chemicals - Health Effects.
61 - Drinking Water System Components - Health Effects.

(Application for copies should be addressed to the National Sanitation Foundation, 3475 Plymouth Road, P.O. Box 1468, Ann Arbor, MI 48106.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Materials.

3.2.1 Cartridge. The containment cartridge, type I and II, and any components of the cartridge which come in contact with the process water shall meet the requirements of NSF 61, section 4. The cartridge and containment screens (see 3.4) shall be made of high density polyethylene in accordance with ASTM D 1248, type III or IV.

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3.2.1.1 Type I gasket seals. A square cut gasket seal, sized to seal type I canisters, shall be provided with each type I cartridge. The gasket seal shall have an outside diameter of 4-1/4 inches, an inside diameter of 4-1/8 inches, and a thickness of 1/8 inch.

3.2.2 Ion exchange resin. The ion exchange resin shall be a trimethyl-benzylammonium resin. The resin and container material (cartridge) shall be certified to NSF 61 or be certified equivalent by an independent EPA approved testing laboratory. The cartridges and manufacturer shall be included in the NSF listing services or be included in an equivalent listing service by an EPA approved independent testing laboratory.

3.2.3 Elemental bromine. The elemental bromine placed into the ion exchange resin shall be certified to NSF 60 or be certified equivalent by an independent EPA approved testing laboratory. The bromine and manufacturer shall be included in the NSF listing services or be included in an equivalent listing service by an EPA approved testing laboratory.

3.2.4 Hazardous material. Hazardous items are substances, mixtures, materials, components or equipment that may cause personal injury, property damage or environmental deterioration through transportation, use or disposal. These items shall be marked in accordance with the requirements **of** public law and regulations. The marking shall include as applicable: Name of product; quantity warning symbol; signal word designating degree of hazard; affirmative statement of hazards; precautionary measures covering actions to be followed or avoided instructions in case of contact or exposure; antidotes and notes to physicians; instructions in case of fire, spill, or leakage; instructions for handling and storage; and disposal instructions. Characteristics and operating hazards that require labeling include: toxic, high toxic, irritant, corrosive, strong sensitizer, combustible liquid, flammable, extremely flammable liquid, dangerously reactive, pressure-generation, explosive, magnetic, ionizing radiation, non-ionizing radiation, high voltage, implosion, noise, and etiologic agent. Marking of hazardous materials shall be in accordance with 5.4.

3.2.5 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term “recovered materials” means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2.6 Prohibited material. The following materials are prohibited from use in bromine cartridges covered by this specification:

(a) Adhesives - Adhesives or adhesive tape shall not be used in the fabrication or assembly of the cartridge.
(b) Asbestos.
(c) Known or suspected carcinogens.

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3.2.7 Mercury. The bromine cartridge shall be free of mercury contamination. During the manufacturing process, tests, and examination, the product to be offered for acceptance shall not come in direct contact with mercury or any of its components , nor with any mercury containing device employing a single boundary of containment (see 6.3).

3.3 Toxic products . The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency.

3.3.1 Material safety data sheet (MSDS). The contracting activity shall be provided a material safety data sheet at the same time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 6.6).

3.3.2 EPA registration. Cartridges shall receive Environmental Protection Agency registration in accordance with EPA 40 CFR, part 22.

3.4 Construction. Construction shall be as specified in 3.4.1 and 3.4.2 (see 6.3). The cartridge shall be of the physical dimensions specified herein. No element (internal or external) of the type I or type II cartridge shall be assembled by any form of compression fitting (including shrink fit, press fit, and spin weld).

3.4.1 Type I cartridge. The type I bromine cartridge and associated components shall conform to the physical specifications of figure 1.

3.4.2 Type II cartridge. The type II bromine cartridge and associated components shall conform to the physical specifications of figure 2.

3.5 Performance. Cartridges provided under this specification for bromination of potable water shall meet the performance requirements of 3.5.1 through 3.5.7. The characteristics and corresponding test procedures are summarized in table I.

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TABLE I. Cartridge performance requirements.

Property	Requirement		Test method
	Type I	Type II	
Process water flow rate, gal/min	2	10	4.3.2.1.1
Resin dryness (screening test)	Free flowing	Free flowing	4.4.4
Resin dryness (verification test), percent by weight	≤ 9.0	≤ 9.0	4.3.2.2.1
Resin weight, pounds (kilograms)	2.75 - 3.00 (1.25 - 1.36)	9.75 - 10.00 (4.43 - 4.54)	4.3.2.2.1 and 4.4.3
Bromine content, percent by weight	30 ± 3	> 30 ± 3	4.3.2.2.2 and appendix B
Pressure drop, lb/in^2	< 10	< 12	4.3.2.1.2
Effluent bromine concentration, p/m	> 10	> 10	4.3.2.1.3 and appendix A
Shelf life, years	≥ 2	≥ 2	

3.5.1 Process water flow rate.

3.5.1.1 Type I cartridge. Type I cartridges shall disinfect potable water as specified in 3.5.3 at a process flow rate of 2 gallons per minute (gal/min).

3.5.1.2 Type II cartridge. Type II cartridges shall disinfect potable water as specified in 3.5.3 at a process water flow rate of 10 gal/min.

3.5.2.1 Type I cartridge. Type I cartridge shall have a pressure drop of not greater than 10 pounds per square inch (lb/in^2) (69 kilopascals (kPa)) at the process water flow rate specified in 3.5.1.1.

3.5.2.2 Type II cartridge. Type II cartridge shall have a pressure drop of not greater than 12 lb/in^2 (83 kPa) at the process water flow rate specified in 3.5.1.2.

3.5.3 Effluent bromine concentration. Bromine concentration in the process water exiting both the type I and type II cartridges shall be greater than 10 parts per million (p/m) for 6 hours of continuous operation at the flow rate specified in 3.5.2.1 and 3.5.2.2.

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3.5.4 Effluent water quality. The contaminant concentrations in the water exiting the cartridge shall not exceed the limits for the contaminants listed in the EPA 40 CFR, part 141 and part 143, for inorganic and organic contaminants (see 6.8).

3.5.5 Resin weight.

3.5.5.1 Type I cartridge. Type I cartridge shall contain 2.75 to 3.00 pounds (1.25 to 1.36 kilograms (kg)) of dried ion exchange resin impregnated with bromine.

3.5.5.2 Type II cartridge. Type II cartridge shall contain 9.75 to 10.00 pounds (4.43 to 4.54 kg) of dried ion exchange resin impregnated with bromine.

3.5.6 Elemental bromine content. Elemental bromine content of both the type I and type II cartridges shall be 30 $\pm$ 3 percent of the total weight of the ion exchange resin specified for each cartridge.

3.5.7 Shelf life. Shelf life of both the type I and type II cartridges shall be not less than 2 years at a recommended storage temperature of 82 degrees Fahrenheit (“F) or cooler (see 6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibilitv for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the tests as specified in 4.3.2 (see 6.3 and 6.7).

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4.3.1 Sampling for first article inspection. A sample shall consist of four cartridges and two samples shall be selected at random from each lot. These samples shall be designated as sample A and sample B.

4.3.1.1 Lot. A lot shall consist of all cartridges produced from one batch of bromine impregnated ion exchange resin, produced in one facility, using the same materials and processes.

4.3.2 First article inspection tests.

4.3.2.1 Performance tests. These tests shall be performed on each cartridge of sample A.

4.3.2.1.1 Process water flow rate. Process water flow rate shall be maintained during the test procedure specified in 4.3.2.1.3.

4.3.2.1,2 Pressure drop. The pressure drop shall be determined during tests for effluent bromine concentration (see 4.3.2.1.3). The pressure drop, from inlet to outlet of canister-cartridge assembly, shall be measured at points within 2 inches of the inlet and outlet of the assembly.

4.3.2.1.3 Effluent bromine concentration. Water in accordance with ASTM D 1193, type IV quality or better shall be used for this test. A water flow rate, as specified in 3.5.1, shall be maintained through the cartridge for a period of 6 hours at a feed water temperature of 86"F. Effluent samples shall be taken of the initial effluent (no more than 5 minutes elapsed time) and after each l-hour interval. Each effluent sample shall be analyzed for bromine concentration using the spectrophotometric method described in appendix A or the amperometric titration method of APHA Standard Methods for the Examination of Water and Wastewater. Samples shall be analyzed immediately after collection. None of the samples shall yield bromine concentrations below the level required by table I.

4.3.2.1.4 Effluent water quality. Water in accordance with ASTM D 1193, type IV quality or better shall be used for this test. Analysis for the organic and inorganic materials specified in 3.5.4 shall be performed in accordance with currently accepted EPA methods. If no EPA method is provided, analysis may be performed in accordance with APHA Standard Methods for the Analysis of Water and Waste Water. The following samples shall be collected during this test: an initial effluent sample (5 minutes elapsed time); an effluent sample after 3 hours elapsed time; an effluent sample after 6 hours elapsed time; and three influent samples, one taken with each effluent sample. In addition, one sample of the water entering the cartridge shall be taken at the beginning of the same test run and analyzed in conjunction with the three effluent samples. Levels of a specific contaminant present in the influent water shall be subtracted from the contaminant levels present in the effluent water sample and the result compared to the requirements of 3.5.4.

4.3.2.2 Resin characteristics tests. These tests shall be performed on each cartridge of sample B.

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4.3.2.2.1 Resin weight. Using a balance with minimum precision of plus or minus 0.01 lb (+ 5g), weigh each sample cartridge. The cartridge shall then be opened, the resin shall be removed, and the cartridge shall be reweighed (and any pieces of the cartridge produced in opening). At least 50 milliliters (mL) of bromine impregnated resin shall be retained for determining resin dryness (see 4.3.2.2.3). The weight of bromine impregnated resin shall be determined by subtracting the empty cartridge weight from the full cartridge weight.

4.3.2.2.2 Elemental bromine content. The elemental bromine content shall be determined as specified in appendix B.

4.3.2.2.3 Resin dryness. The tests specified in 4.3.2.2.3.l and 4.3.2.2.3.2 shall be performed to determine resin dryness. The dryness verification test of 4.3.2.2.3.2 shall be performed only if the resin passes the dryness screening test of 4.3.2.2.3.1.

4.3.2.2.3.1 Resin dryness screening test. Resin from the cartridges shall be tested for dryness as follows. At least 50 mL of the resin shall be poured into a 60-degree angle glass funnel with a 6 inch (15.25 cm) long stem. The inside diameter of the stem shall be 0.138 inches (3.5 mm). The resin shall flow freely through the funnel. The resin exiting the funnel shall be collected for dryness verification tests as specified in 4.3.2.2.3.2.

4.3.2.2.3.2 Resin dryness verification test. The resin collected following the dryness screening test shall be tested as specified in appendix C. The weight loss shall not exceed 9.0 percent.

4.4 Quality conformance inspection. Sample bromine cartridges (see 4.4.1) shall be subjected to the examinations and tests specified in 4.4.2 through 4.4.4 (see 6.3 and 6.7).

4.4.1 Sample size for quality conformance inspection. As a minimum, the contractor shall randomly select a sample quantity from each lot of bromine cartridges as specified in table II (see 6.7).

TABLE II. Sample size for quality conformante inspections and tests.

Lot size	Sample size
2 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	10
281 to 500	11
501 to 1200	15
1201 to 3200	18
3201 to 10,000	22

4.4.1.1 Lot. A lot shall consist of all cartridges produced from one batch of bromine impregnated ion exchange resin, produced in one facility, using the same materials and processes, and being offered for delivery at one time.

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4.4.2 Cartridge construction. The sample cartridge shall be examined for defects and for evidence of resin leakage past the containment screen.

4.4.3 Weight of resin. Using a balance with minimum precision of plus or minus 0.01 lb (+5g) , weigh each sample cartridge to determine the full cartridge weight. Weigh an empty sample cartridge to determine the empty cartridge weight (the empty cartridge weight may be obtained from 4.3.2.2.1). The weight of bromine impregnated resin shall be determined by subtracting the empty cartridge weight from the full cartridge weight.

4.4.4 Resin dryness. Resin from the bulk resin container used to fill the cartridges shall be tested for dryness in accordance with the dryness screening test procedures of 4.3.2.2.3.1. Resin which passes the dryness screening test shall then be tested in accordance with the dryness verification test procedures of 4.3.2.2.3.2.

4.5 Toxicological product formulation. The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.5.)

5.1 Presentation. Preservation shall be level A.

5.1.1 Cleaning. Cartridges shall be cleaned in accordance with process Cl of MIL-P-116.

5.1.2 Drying. Cartridges shall be dried in accordance with process D1 of MIL-P-116.

5.1.3 Preservative. No preservative shall be applied to any part of the bromine cartridge.

5.2 Unit packs. Each cartridge shall be individually packaged (see 6.9) as follows:

(a) A polyethylene cap shall be installed on the cartridge outlet nipple threads.
(b) The cartridge shall be enclosed in a heat sealable, laminate film consisting of 1 layer of 0.5-mil thick, oriented polyester and 1 layer of 0.5-mil thick, heat sealable polyethylene containing no slip additives. The film shall have the properties listed in table III.

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during the life of the commutator. Parts subject to wear or deterioration shall

TABLE III. Packaging film requirements.

Property	Typical value
Total thickness	0.0025 inches
Tensile strength M.D. break	15.0 pounds/inch of width
Burst strength (Mullen)	45.0 lb/in^2
Water vapor transmission rate	0,7 gin/loo square inches/24 hours

(c) The polyester-sealed type I cartridge shall be placed in a 40-pound, fiberboard, polyethylene-coated inner and outer surface tube. The paper board cylinder for type I cartridges shall be 14.5 inches long, 4.5 inches inside diameter, and 4.75 inches outside diameter.

(d) The polyester-sealed type II cartridge shall be placed in a 40-pound, fiberboard, polyethylene-coated inner and outer surface tube. The paper board cylinder for type II dimensions shall be 16.5 inches long, 8.5 inches inside diameter and 8.75 inches outside diameter.

(e) A gasket conforming to requirements of 3.2.1.1 shall be placed in the paper board cylinder on top of the sealed cartridge.

(f) Polyethylene end caps are secured to the paper board cylinder with waterproof tape in accordance with PPP-T-60.

(g) The paper board cylinder shall be enclosed in a 4-mil thickness polyethylene bag tied at the open end with polymer-coated wire tie ribbon.

5.3 Packing level. Type I cartridges shall be packed at level C, 8 cartridges to a carton. Type II cartridges shall be packed at level C, 10 cartridges to a carton. The carton shall be a weather-resistant fiberboard box in accordance with PPP-B-636.

5.4 Marking. In addition to any special markings, cartridges preserved as above shall be marked in accordance with MIL-STD-129 on paper board cylinder and shipping container.

5.4.1 Special markings. In addition to MIL-STD-129 markings, each paper board cylinder and shipping container shall have the following affixed:

(a) Manufacturer name and address.
(b) National Stock Number.
(c) Statement of where cartridge is used.

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(d) Chemical content.
(e) EPA registration number.
(f) Health warnings.
(g) First aid.
(h) Net weight.
(i) Disposal method.
(j) Lot number.
(k) Manufacturing date.
(1) Expiration date.
(m) Manufacturer part number (including modification number).
(n) National Sanitation Foundation (NSF) listing mark (or evidence of an equivalent listing).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Bromine cartridges covered by this specification are intended for use in brominators aboard naval surface ships for the disinfection of potable water.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Type of cartridge required (see 1.2).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Whether first article inspection is required (see 3.1).

6.2.1 Bid data. The following information should be submitted with each bid:

(a) Manufacturer's model and type designation of cartridge.
(b) EPA product registration number.
(c) Certification of compliance with FDA and NSF requirements as specified herein.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.2.7 and 3.5.7	DI-MISC-80678	Certification/data report	10.3.1 does not apply
3.4	DI-DRPR-81001	Conceptual design drawings and associated lists	----
4.3 and 4.4	DI-QCIC-81110	Inspection and test plan	----
4.3 and 4.4	UDI-T-23732	Procedures, tests	..-.
4.4	DI-NDTI-80809	Test/inspection reports	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first Production item, a sample selected from the first ___ production items, a standard production item from the contractor's current inventory (see 3,1), and the number of items to be tested as specified in 4.3, The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations) approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Goverment reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Materials Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.7 Lot acceptance and rejection criteria. Failure of any single cartridge to pass any test should be cause for rejection of the entire lot. For quality conformance inspection, the contractor has the option of screening 100 percent of the rejected lot for the defective characteristics or providing a new lot, which has been tested in accordance with the sampling plan herein. The contractor should maintain for a period of 3 years after contract completion, records of inspections, tests, and any resulting rejections.

6.8 Effluent water qualitv. The requirements (see 3.5.4) are the same as those specified in the Manual of Naval Preventive Medicine, Chapter S, Water Quality Ashore, Appendix F, Sections F-1, F-2, and F-3.

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6.9 Alternate packaging. Alternate packaging methods may be submitted by the manufacturer and are subject to review and approval by the contracting activity.

6.10 Patent notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related there to.

6.11 Subject term (key word) listing.

Brominator
Trimethylbenzylammonim (ion-exchange) resin

Preparing activity:
Navy - SH
(Project 4610-N116)

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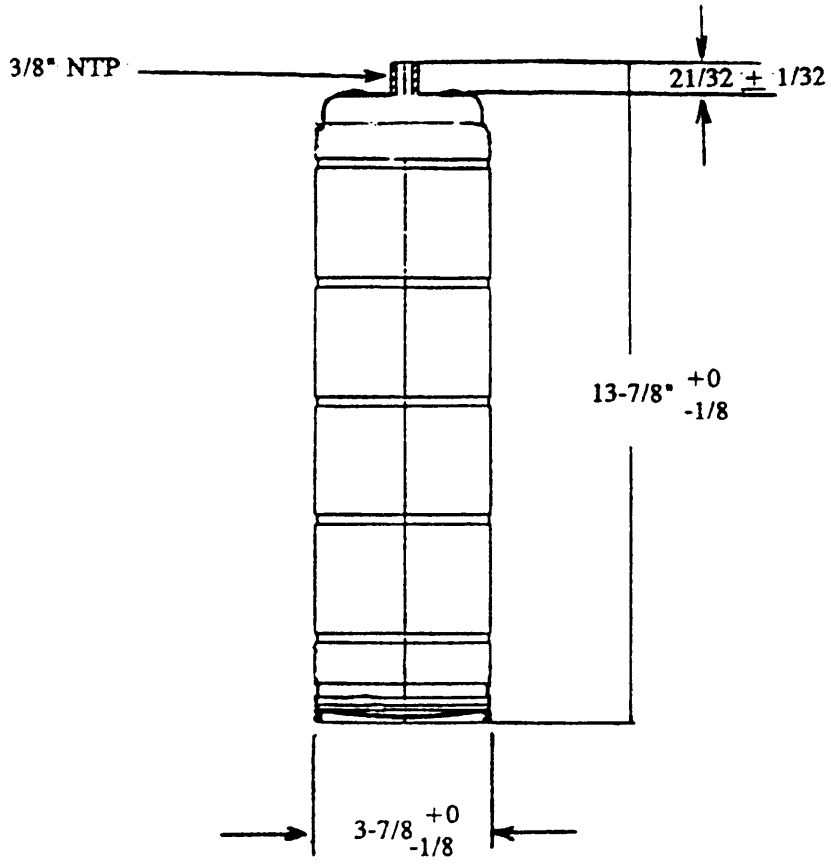


FIGURE 1. The I bromine cartridge.

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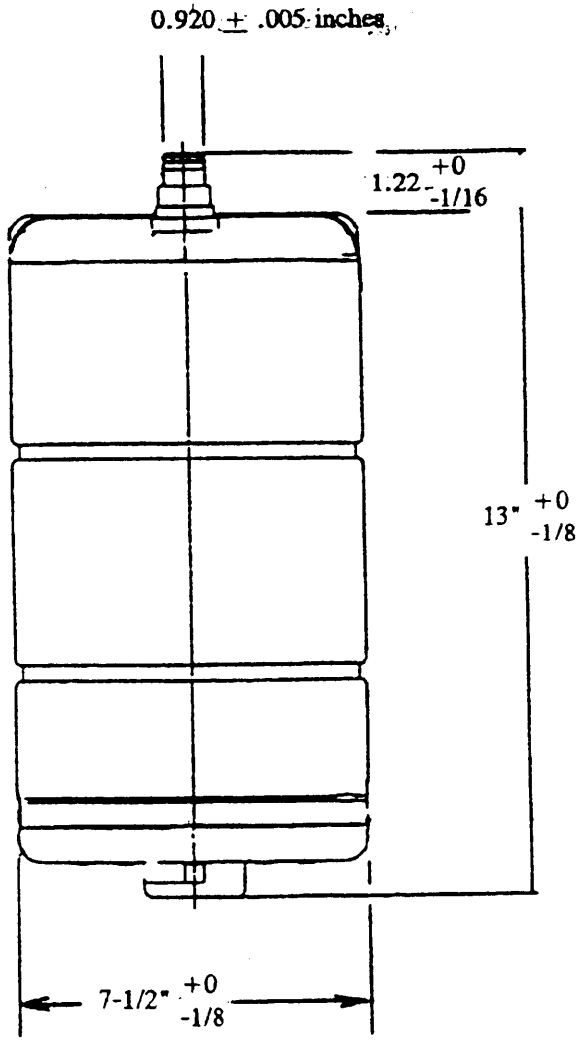


FIGURE 2. Type II bromine cartridge.

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APPENDIX A

BROMINE CONCENTRATION IN WATER

10. SCOPE

10.1 Scope. This method is intended for the determination of bromine concentration in water. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

10.2 Summary of method. A reagent is added to a sample producing color whose intensity is proportional to the concentration of bromine. The absorbance of the red color is measured at 530 nanometers (rim) wavelength and compared with a standard calibration to determine the bromine concentration.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. APPARATUS

30.1 Spectrophotometer and cuvet(s) for measuring absorbance at a wavelength of 530 nm.

30.2 Stoppered volumetric flasks (four 100-mL and one 1-L capacity are suitable).

30.3 Glass pipettes (1, 2, 3, and 5 mL capacity are suitable).

30.4 Analytical balance with precision of plus or minus 0.0001 g.

40. REAGENTS

40.1 N, N-diethyl-p-phenylenediamine (DPD). (This reagent is available in 0.25 g portions sealed in plastic “pillows” from Hach Co., P.O. Box 389, Loveland, CO 80539.)

40.2 Bromine, ACS reagent grade.

50. PROCEDURE

50.1 Standards. Using the deionized water to be used as influent for the cartridge, prepare bromine standards of approximately 5, 10, 20, and 30 p/m. First a concentrated (approximately 1000 p/m) solution is prepared to be used for dilution to the desired concentration range, A stoppered 100-mL volumetric containing approximately 50 mL of deionized water is weighed to the nearest 0.0001 g. Approximately 0.1 g (0.03 mL) of bromine is added to the water in the weighed 100-mL flask, then stoppered and reweighed to determine the exact weight of bromine added. Water is then added to bring the volume to 100 mL. The stoppered flask is shaken and placed in a refrigerator, allowing the bromine to

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completely dissolve. As soon as the bromine has completely dissolved the dilute solutions (approximately 5, 10, 20, and 30 p/m) are prepared by appropriate dilution using glass pipettes and volumetric flasks (for example, dilute 1 mL of the 1000-p/m solution to 100 mL with deionized water to give a 10-p/m standard).

50.2 Calibration. With the spectrophotometer set for a wavelength of 530 nm, place a cuvet containing deionized water in the sample beam and adjust the absorbance to read 0. Take a 25-mL sample of a freshly Prepared standard and 0.25 g DPD (contents of a DPD pillow). Mix well for 30 seconds, fill the cuvet and record the maximum absorbance reading immediately (the color intensity fades with time) . Continue the same procedure for the other standards using the same or matched cuvets, Construct a calibration curve of absorbance versus bromine concentration.

50.3 Determination. Add 0.25 g DPD reagent *to* a 25-mL sample of brominated water whose concentration you wish to determine. (Water samples must be tested for bromine as soon as collected.) Mix the DPD reagent and sample for 30 seconds. Using the same cuvet(s) used to obtain absorbance readings of the calibration standards measure the absorbance. If the absorbance reading exceeds the range of the calibration tune, take another portion of the sample, dilute it such that it is within the calibration range, add DPD reagent to 25 mL of the diluted sample and measure the absorbance. Multiply the concentration value obtained from the calibration tune by the dilution factor, if dilution was necessary.

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APPENDIX B

TEST FOR DETERMINATION OF WEIGHT PERCENT BROMINE OF ION EXCHANGE RESIN

10. SCOPE

10.1 Scope. This method is intended for the determination of weight percent bromine in brominated ion exchange resins. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

10.2 Summary of method. Bromine shall be extracted from a weighed portion of ion exchange resin with 4 percent HC1. The extracted bromine solution shall be neutralized and the concentration of bromine in a measured volume is determined by the method of appendix A or equivalent method. From the total amount of bromine extracted and the weight of the sample, the percent of bromine shall be calculated.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. APPARATUS

30.1 This method requires the apparatus called for in appendix A plus the following.

30.1.1 Plastic weigh dish (small).

30.1.2 Liquid chromatography column (10 mm ID, 300 mm long), Pyrex brand tube with a glass fitted disc and TFE stopcock plug.

30.1.3 Addition funnel, 500-mL or 1000-mL capacity.

30.1.4 Small funnel.

30.1.5 Ring stand.

30.1.6 Buret holder.

40. REAGENTS

40.1 As needed for appendix A.

40.2 Four percent hydrochloric acid (HC1).

40.3 One normal (lN) sulfuric acid (H_2SO_4).

40.4 One normal (1N) sodium hydroxide (NaOH).

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50. PROCEDURE

50.1 Tare a weigh dish. Place approximately 1.0 g of a representative sample of brominated resin in the tared weigh dish. Record weight.

50.2 Secure a liquid chromatography (LC) column above a 1000-mL volumetric flask so that the column effluent will be collected in the flask. Secure an addition funnel above the column filled with 4 percent HC1.

50.3 Carefully pour the dry brominated resin into the LC column filled with 4 percent HC1.

50.4 Start feeding HC1 from the funnel at a rate sufficient to maintain liquid above the top of the resin layer. Adjust the effluent column flow rate to 10 $\pm$ 2 mL per minute and continue the extraction until a total of 1000 mL is collected in the volumetric flask. Stopper and invert the flask several times to assure a homogeneous solution. Take 25 mL of the solution and adjust (if necessary) the pH to between 6 and 7 using 1 N sodium hydroxide or sulfuric acid. Make appropriate dilutions for analysis.

50.5 Determine the bromine concentration of the diluted solution following the procedures of appendix A, making further dilutions when necessary to obtain an absorbance value within the calibrated range. From the total concentration of all three 1000-mL samples collected, calculate the weight of bromine extracted and the percent bromine in the fresh resin.

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APPENDIX c

RESIN DRYNESS
VERIFICATION TEST PROCEDURES

10. SCOPE

10.1 Scope. This method is intended for the determination of the weight loss by an ion exchange resin when exposed to elevated temperatures. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

10.2 Summary of method. A sample of ion exchange resin is exposed to an elevated temperature for a specified period of time. The weight loss of the sample is determined gravimetrically and converted to a percent weight loss.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. APPARATUS

30.1 Analytical balance with precision of plus or minus 0.0001g.

30.2 Laboratory convection oven with upper temperature limit greater than 60 degrees Celsius (“C) and constant temperature precision of plus or minus 1.O°C.

30,3 Plastic weighing dish (small).

40. PROCEDURE

40.1 Set convection oven at 60”C. Record oven temperature at 10-minute intervals until temperature remains at 60 ± l°C for three consecutive recordings.

40.2 Tare a weigh dish. Place approximately 1.0 g of a representative sample of brominated resin in the tared weigh dish and reweigh. Record weight.

40.3 Place the weigh dish containing bromine resin in the convection oven for 30 ± 1 minutes. Remove weigh dish from oven and weigh immediately. Record weight.

50. CALCULATIONS

50.1 Percent weight loss. Subtract the weight recorded in 40.3 from weight recorded in 40.2. Divide this result by the weight recorded in 40.2 and multiply by 100. Record result.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-24770(SH)	1992 MARCH 20

3. DOCUMENT TITLE
Cartridges Bromine, For Potable Water Disinfection

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Technical Engineer: r. Tom Marino, NAVSEA 56Y32	(1) Commercial (703) 602-6830	(2) AUTOVON 332-6830
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command (55Z3) Department of the Navy Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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